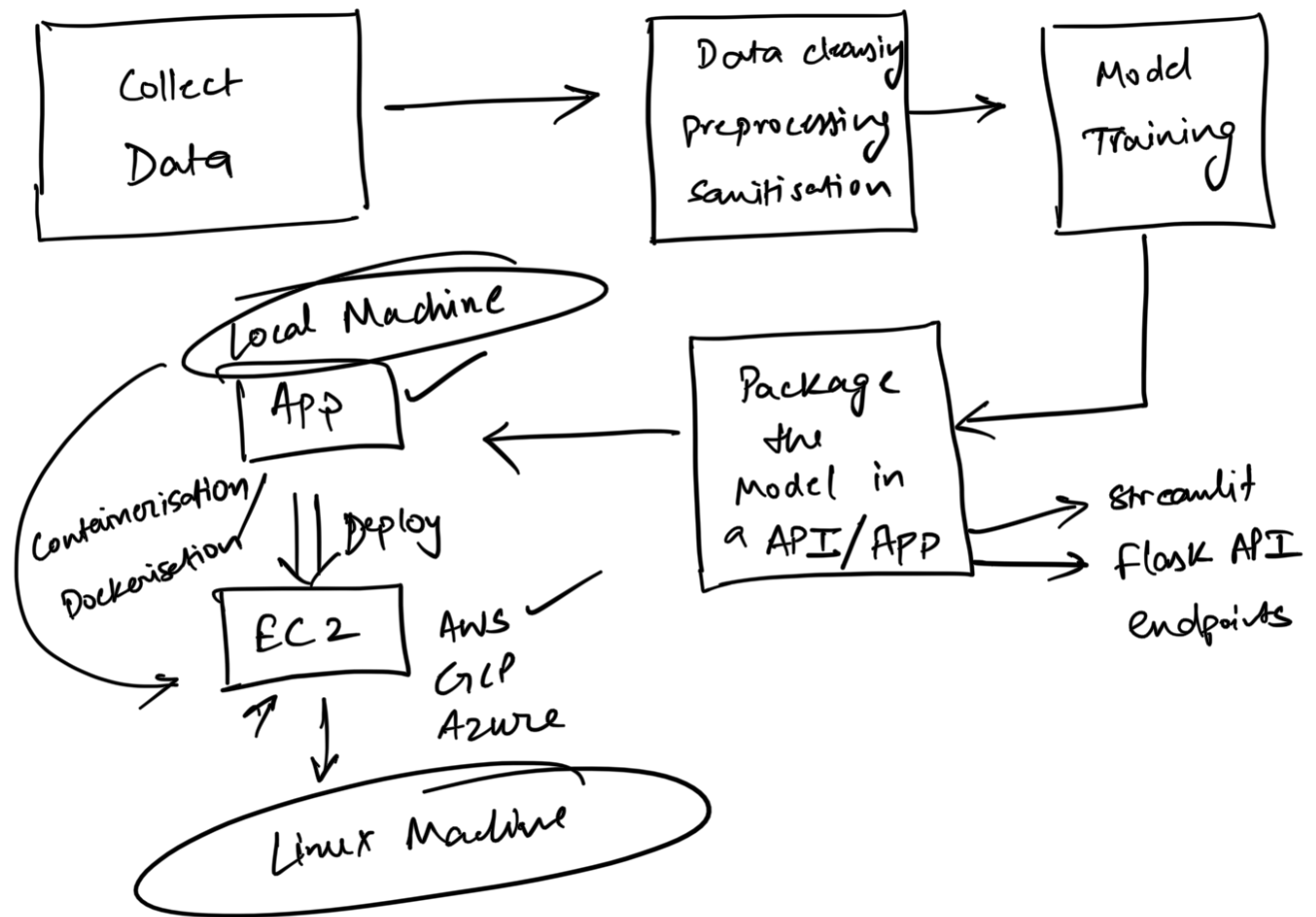


Introduction to Docker

Q: How to run projects/apps on any machine/OS with just a few minimal setup?



Q: What does Deployment entail?

1. Finding a cloud-hosted machine or server
2. Setting up an OS
3. Copy all the files from my local to this server.
4. Install all dependencies. ← python/virtualenv

5. Run the app $\rightarrow$ localhost:5000

IP addresses of the router

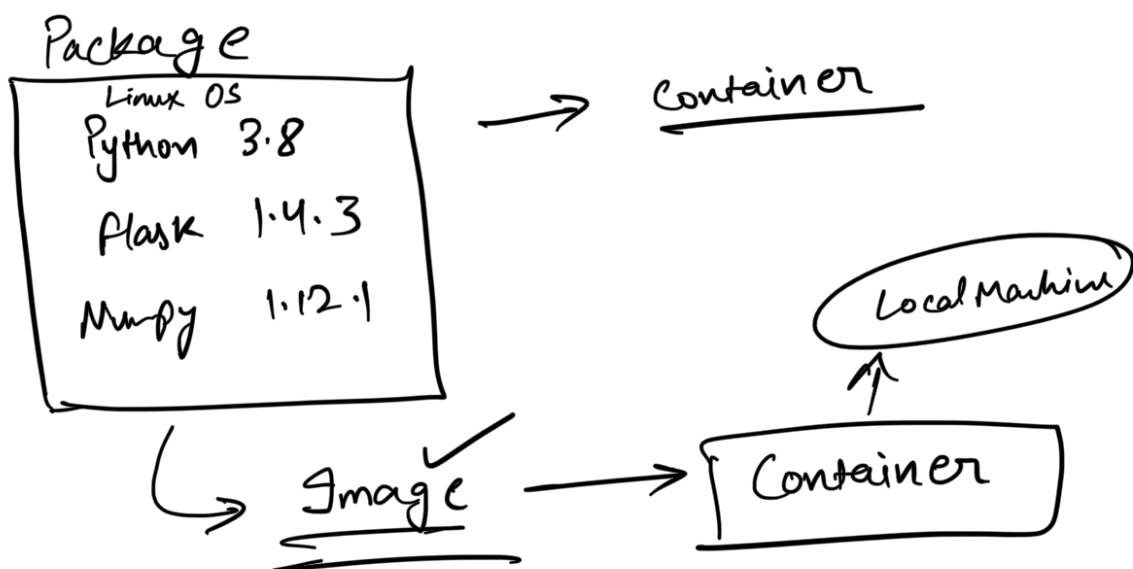
Reasons for failure

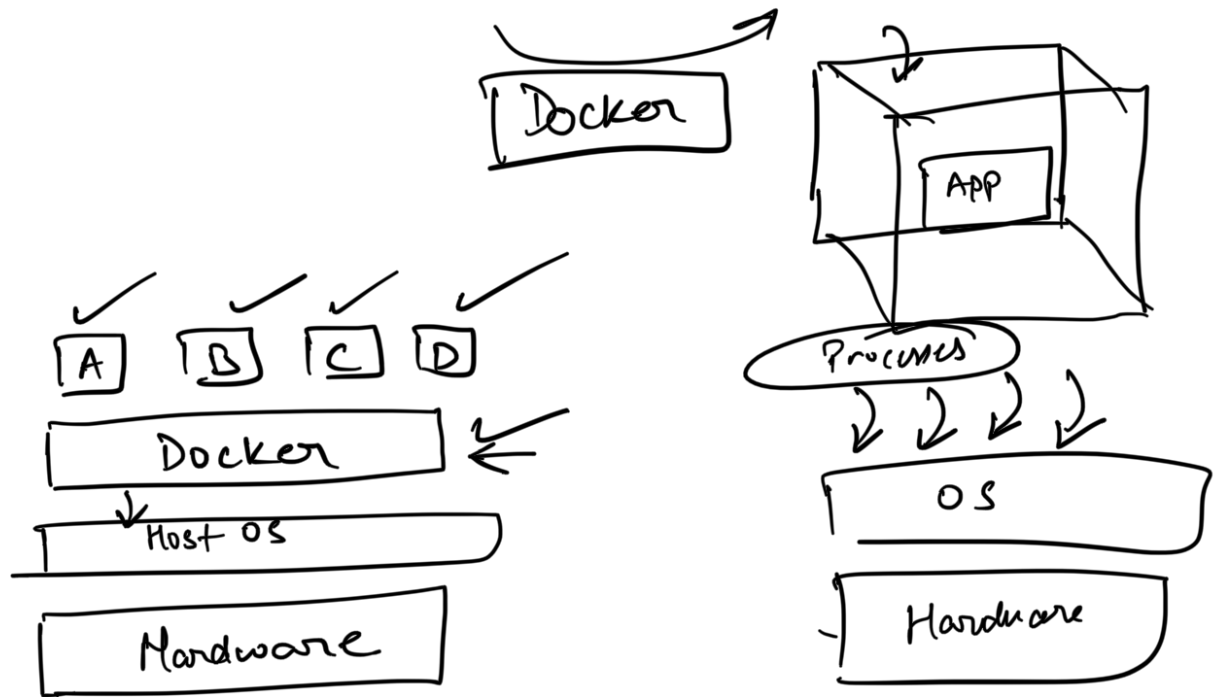
- ① One or more missing files/drivers/dependencies
- ② Version mismatch
- ③ Different OS & configurations

Docker

What is Docker?

A platform for building, running & shipping applications consistently without fail.





Virtual Machine

- An abstraction of a machine (Physical H/W)
- Each VM is a full-blown copy of an OS.
- It takes time to load.
- It takes a lot of disk space from the host storage.

Solutions:

- Lightweight (abstraction at application layer)
- Spins up quickly
- Should be able to run multiple apps.

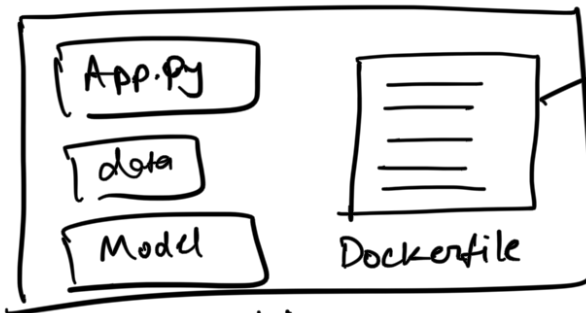
Container



5-10 mins

Docker

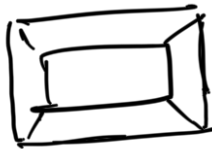
Workflow:



A plain text file that contains instructions which Docker uses to package your app into an image.

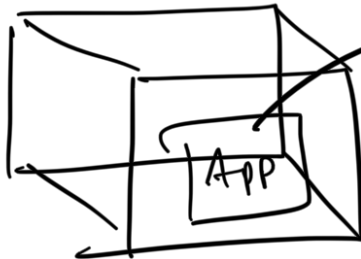
Build

Read-only template containing instructions



Image

Run



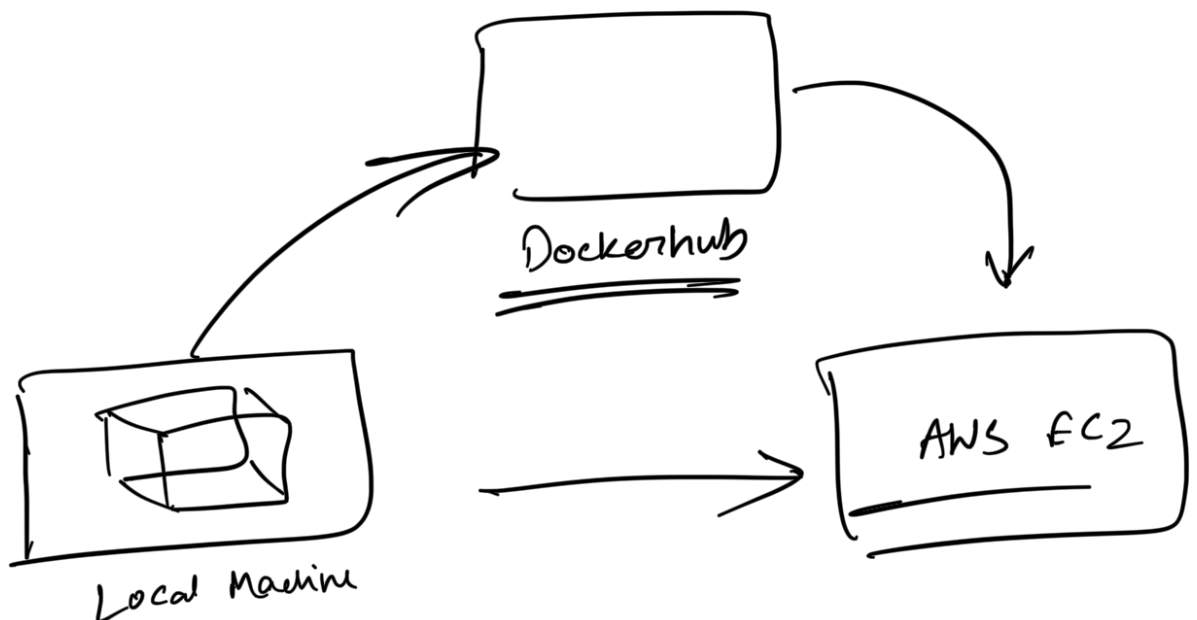
Container

→ Runnable process

localhost: 8000

An Image

- ✓ 1. Install OS
- ✓ 2. A runtime environment → python
3. Copy files. ✓
4. Install all packages from Requirements.txt
5. Run the App



→ AWS Deployment

→ CI/CD (Test, ✓) → AWS ✓

→ MLflow

→ ML System Design -1, 2, 3

→ Sagemaker Pipeline

→ TF ✓ Capstone

